

Exercise Sheet 3

Keep the Peak, Model Only $p(y | x)$

Probabilistic Machine Learning

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3.A Keep the Peak

Exercise 3.A.1: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$.
Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

Exercise 3.A.1: Peak, Mean and a Prior's Peak — Solution

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1. $\log p(D \mid \theta) + \log p(\theta) = (h + a - 1) \log \theta + (n - h + b - 1) \log(1 - \theta) + \text{const.}$ Setting the derivative $\frac{h+a-1}{\theta} - \frac{n-h+b-1}{1-\theta}$ to zero gives the formula — the MLE of a dataset with $a - 1$ extra yeses and $b - 1$ extra noes.
2. MLE $0/4 = 0$: "nobody ever returns". MAP $\frac{0+1}{4+2} = \frac{1}{6} \approx 0.17$. Posterior $\text{Beta}(2, 6)$, mean $\frac{2}{8} = 0.25$. Report the posterior (mean and interval): four users are too few for a point, and the MLE claims certainty the data cannot give.
3. $\frac{h+a-1}{n+a+b-2} = \frac{h+1}{n+2}$ for all h, n needs $a = b = 2$: the MAP under $\text{Beta}(2, 2)$ equals the posterior mean under $\text{Beta}(1, 1)$. "Add one yes and one no" can be read either way — Lecture 04 calls it Laplace smoothing.
4. The CDF is θ^4 , so the interval is $[0.025^{1/4}, 0.975^{1/4}] \approx [0.40, 0.99]$ — the peak at 1 lies at the edge of it.

Exercise 3.A.2: Four Flats and a Villa

Model $y = w_0 + w_1x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; x in m², y in k€.

x	40	60	80	100
y	150	200	280	330

1. Write down $\ell(w_0, w_1, \sigma^2)$ and show that for **every** σ^2 the maximum over (w_0, w_1) is the least-squares fit.
2. Compute $\hat{w}_1 = S_{xy}/S_{xx}$ and $\hat{w}_0 = \bar{y} - \hat{w}_1\bar{x}$, the residuals and the RSS.
3. Compute $\hat{\sigma}_{\text{ML}}^2 = \text{RSS}/n$ and the unbiased $\text{RSS}/(n - 2)$. By which factor does the ML version underestimate σ^2 on average?
4. Give the plug-in 95% band $\hat{y} \pm 1.96\hat{\sigma}_{\text{ML}}$ at 80 m² and at 600 m². Then use the half-width that counts the uncertainty about $\mathbf{w}$, $1.96\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}}$. What do you conclude?

Exercise 3.A.2: Four Flats and a Villa — Solution

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1. $\ell = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - w_0 - w_1 x_i)^2$. The first term has no w , and $\frac{1}{2\sigma^2} > 0$, so the arg max over w is the arg min of the RSS, whatever σ^2 is.
2. $\bar{x} = 70, \bar{y} = 240, S_{xx} = 2000, S_{xy} = 6200$: $\hat{w}_1 = 3.1$ k€ per m², $\hat{w}_0 = 240 - 217 = 23$. Fitted 147, 209, 271, 333; residuals 3, -9, 9, -3; RSS = 180.
3. $\hat{\sigma}_{\text{ML}}^2 = 45$ ($\hat{\sigma} \approx 6.7$); $\text{RSS}/2 = 90$ (≈ 9.5). Since $E[\text{RSS}] = (n - 2)\sigma^2$, the ML version is on average $\frac{n-2}{n} = \frac{1}{2}$ of σ^2 — with 4 points and 2 weights, half the noise disappears into the fit.
4. Plug-in: 271 ± 13 at 80 m², 1883 ± 13 at 600 m². With the uncertainty about $\mathbf{w}$: factor $\sqrt{1.3} \approx 1.14$ at 80 m² (± 15), $\sqrt{141.7} \approx 11.9$ at 600 m² (± 157). The plug-in band is as narrow at the villa as at a flat; the honest one is twelve times wider — and even that one trusts the straight line out to 600 m².

3.B Classification: Only $p(y | x)$

Exercise 3.B.1: Cross-Entropy and Its Gradient

$h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$ with $\sigma(z) = 1/(1 + e^{-z})$ and $\ell(\mathbf{w}) = \sum_i [y_i \ln h_i + (1 - y_i) \ln(1 - h_i)]$.

1. Show $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.
2. Show $\frac{\partial \ell}{\partial \mathbf{w}} = \sum_i (y_i - h_i) \mathbf{x}_i$.
3. Six transactions, $\mathbf{x} = (1, \text{new device})$: new device — fraud, fraud, legitimate; known device — fraud, legitimate, legitimate. Compute the gradient at $\mathbf{w} = 0$.
4. Set both components of the gradient to zero: which scores does the MLE give to new and to known devices? Give $\hat{w}_0, \hat{w}_1$ and read $\hat{w}_1$ as an odds ratio.

Exercise 3.B.1: Cross-Entropy and Its Gradient — Solution

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1. $\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)(1 - \sigma(z)).$
2. $\frac{\partial}{\partial \mathbf{w}} [y \ln h + (1 - y) \ln(1 - h)] = \left(\frac{y}{h} - \frac{1-y}{1-h}\right) h(1 - h)\mathbf{x} = (y - h)\mathbf{x}$; sum over i . Error $\times$ feature, as in LMS.
3. At $\mathbf{w} = 0$ every $h_i = 0.5$. Intercept: $\sum_i (y_i - 0.5) = 3 - 3 = 0$. Slope: only new devices count, $0.5 + 0.5 - 0.5 = 0.5$. Gradient $(0, 0.5)$: raise the weight of "new device".
4. Slope component: $\sum_{\text{new}} (y_i - h_{\text{new}}) = 2 - 3h_{\text{new}} = 0$, so $h_{\text{new}} = 2/3$; then the intercept component gives $h_{\text{known}} = 1/3$ — each group's observed fraud rate. $\hat{w}_0 = \ln \frac{1/3}{2/3} = \ln \frac{1}{2} \approx -0.69$, $\hat{w}_1 = \ln 2 - \ln \frac{1}{2} = \ln 4 \approx 1.39$: a new device multiplies the odds of fraud by 4.

Exercise 3.B.2: The Peak That Runs Away (Python)

```
import numpy as np
x = np.array([-2.0, -1.5, -1.1, -0.6, -0.3, 0.2, 0.6, 1.0, 1.4, 1.9])
y = np.array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1])
sigma = lambda z: 1 / (1 + np.exp(-z))

def fit(tau2=None, steps=10_000, eta=0.05):
    w0 = w1 = 0.0
    for t in range(1, steps + 1):
        h = sigma(w0 + w1 * x)
        g0 = ___ # gradient for the intercept
        g1 = ___ # gradient for the slope
        if tau2 is not None:
            g1 -= ___ # log-prior N(0, tau2) on the slope
        w0, w1 = w0 + eta * g0, w1 + eta * g1
        if t in (10, 100, 1_000, 10_000):
            h = sigma(w0 + w1 * x)
            print(t, round(w1, 2), np.sum(y * np.log(h) + (1 - y) * np.log(1 - h)))
```

1. Fill in the blanks and run `fit()`. What happens to the slope, the log-likelihood and the scores? Why will it never stop?
2. Run `fit(tau2=1)` and `fit(tau2=100, steps=100_000)`. Explain the difference with the MAP formula of the lecture.

Exercise 3.B.2: The Peak That Runs Away — Solution

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```
g0 = np.sum(y - h)
g1 = np.sum((y - h) * x)
g1 -= w1 / tau2
```

1. Slope $1.37 \rightarrow 3.64 \rightarrow 8.32 \rightarrow 17.76$; log-likelihood $-2.68 \rightarrow -1.04 \rightarrow -0.36 \rightarrow -0.06$. The data are separable, so scaling up the weights always raises the likelihood towards 0: the scores approach exactly 0 and 1 and the peak does not exist — three heads in three flips.
2. $\tau^2 = 1$: slope 1.43 from about 100 steps on, log-likelihood -2.54 . The log-prior $-w_1^2/(2\tau^2)$ outgrows the vanishing gain in likelihood, so a finite MAP exists. $\tau^2 = 100$: a weaker prior, slope 7.83 — finite, but steep. $\lambda = 1/\tau^2$ decides how far the peak is allowed to run.

3.C When Is a Score a Probability?

Exercise 3.C.1: The Rebalanced Fraud Model

A model is trained on a 50/50 sample; on real traffic fraud is **1 in 1,000**. Undersampling kept every fraud and a random share of legitimate cases.

1. Use Bayes' theorem in odds form to show: $\text{real odds} = \text{model odds} \times \frac{1/999}{1}$. Which assumption about $p(\mathbf{x} \mid y)$ do you need?
2. Convert the scores 0.5, 0.9 and 0.99 into real fraud probabilities.
3. Blocking a legitimate card costs €10, missing a fraud €500. Above which real probability should a card be blocked, and which **raw score** is that?

Exercise 3.C.1: The Rebalanced Fraud Model — Solution

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1. Posterior odds = $\frac{p(\mathbf{x}|\text{fraud})}{p(\mathbf{x}|\text{legit})} \times \text{prior odds}$. Sampling that depends only on y leaves $p(\mathbf{x} \mid y)$ unchanged, so the likelihood ratio is the same in both worlds; only the prior odds change, from 1 (training) to $1/999$ (real).
2. 0.5: odds $1/999$, $p = 0.10\%$. 0.9: odds $9/999$, $p = 9/1008 \approx 0.89\%$. 0.99: odds $99/999$, $p = 99/1098 \approx 9.0\%$.
3. Block if $500p > 10(1 - p)$, i.e. $p > \frac{10}{510} \approx 1.96\%$ (real odds > 0.02). Raw odds $> 0.02 \cdot 999 = 19.98$, raw score $> 19.98/20.98 \approx 0.95$. A threshold of 0.5 on the raw score would block cards whose real risk is 0.1%.

Exercise 3.C.2: Weight Decay Is a Prior

1. With $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ and a prior $\mathbf{w} \sim \mathcal{N}(0, \tau^2 \mathbf{I})$, show that the MAP minimises $\text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$ with $\lambda = \sigma^2 / \tau^2$.
2. One weight, no intercept, $y = wx + \varepsilon$. Show $\hat{w}_{\text{MAP}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$.
3. Data $(x, y) = (1, 2), (3, 7)$, $\sigma^2 = 1$, $\tau^2 = 0.1$. Compute the MLE and the MAP. What happens as $\tau^2 \rightarrow \infty$ and as $\tau^2 \rightarrow 0$?

Exercise 3.C.2: Weight Decay Is a Prior — Solution

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1. $\log p(\mathbf{w} \mid D) = -\frac{\text{RSS}(\mathbf{w})}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\tau^2} + \text{const.}$ Multiplying by $-2\sigma^2$ turns the maximum into the minimum of $\text{RSS} + \frac{\sigma^2}{\tau^2} \|\mathbf{w}\|^2$.
2. The derivative of $\sum_i (y_i - wx_i)^2 + \lambda w^2$ is $-2 \sum_i x_i y_i + 2w \sum_i x_i^2 + 2\lambda w$; zero at the formula.
3. $\sum x_i y_i = 23$, $\sum x_i^2 = 10$, $\lambda = 10$. MLE $23/10 = 2.3$; MAP $23/20 = 1.15$. $\tau^2 \rightarrow \infty$: $\lambda \rightarrow 0$, the MLE (a flat prior). $\tau^2 \rightarrow 0$: $\lambda \rightarrow \infty$, $\hat{w} \rightarrow 0$ — the prior overrules two data points, as a strong Beta prior did in Lecture 02.